

A CLASS OF DISTRIBUTION-FREE ANALYSIS OF VARIANCE TESTS

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1. Discussion

The development of distribution-free analysis of variance tests has been slow, the main problem being that of testing for effects independently of each other. This problem clearly comes forward in the work of Mood (1950). More recently Lehmann (1963a), (1963b) proposed methods which solve these problems asymptotically. However, since parametric analysis of variance tests are also asymptotically distribution-free, these methods have little justification in practice.

In this paper a simple method of constructing distribution-free analysis of variance tests is given. The rationale of the method is the following. In an analysis of variance one is generally concerned with the testing of contrasts among parameters. The numerators of the corresponding test statistics are then accordingly usually in the form of contrasts among sample means. Furthermore since any contrast among objects can be expressed as a linear sum of paired differences of the objects, the numerators of analysis of variance tests can generally be expressed in terms of paired differences of a set of means. Analysis of variance tests can thus be considered as being built up from the corresponding two-sample test. This suggests a method of constructing distribution-free analysis of variance tests, namely to use as test statistic the same functional form, as in the parametric case, of any two-sample distribution-free test.

The above method is illustrated using Wilcoxon's two-sample test statistic. It is shown that for near alternatives and large sample sizes these tests have the same asymptotic relative efficiency properties, relative to the corresponding parametric tests, as Wilcoxon's test relative to the t-test for two samples. Furthermore these tests have for near alternatives and large sample sizes asymptotically the desirable property of testing for effects independently of each other. However, as mentioned above, these properties are mainly of theoretical interest, and the question is whether these tests are applicable in practice, i.e. when sample sizes are not large and effects not necessarily small. To answer this question the two-way layout with an equal number of observations per cell is considered in detail. Specifically it is found that if interaction effects are not present then row and column effects can be tested, for any sample size, independently of each other. Similar results will clearly hold for complete higher order layouts. This then, although still very limited, is a step nearer to solving the problem satisfactorily.

Clearly other two-sample distribution-free tests can also be used as basis, however, the same problems are likely to crop up.

## 2. Derivation of basic results

Consider the following general analysis of variance situation. Let  $X_{11}, \dots, X_{1n_1}; \dots; X_{m1}, \dots, X_{mn_m}$  be  $m$  random and independent samples from populations with continuous distribution functions  $F_1, \dots, F_m$ . Let  $X_{i.}$  be the mean of the  $i$ th sample and let

$$t_{ij} = X_{i.} - X_{j.},$$

$$v(a, b) = \begin{cases} 1/2 & \text{if } a > b \\ -1/2 & \text{if } a < b, \end{cases}$$

$$V_{ij} = \sum_{h=1}^{n_i} \sum_{k=1}^{n_j} v(X_{ih}, X_{jk}).$$

Consider the alternative

$$(1) \quad F_i(x) = F(x + c_i/\sqrt{N}), \quad i=1, \dots, m,$$

where  $N = \sum_{i=1}^m n_i$  and the  $c$ 's are arbitrary constants.

*Theorem 1.* If, as  $N$  tends to infinity,  $\lim n_i/N$  exists and is bounded away from zero for all  $i$ , then under hypothesis (1) the sets

$$\tilde{t}_{ij} = \sqrt{n_i n_j / \sigma^2 (n_i + n_j)} t_{ij}, \quad i < j,$$

where  $\sigma^2$  is the variance of the variate with distribution function  $F$ , and

$$\tilde{V}_{ij} = \sqrt{12/n_i n_j (n_i + n_j)} V_{ij}, \quad i < j,$$

have, apart from a difference in mean by a constant factor, asymptotically the same multivariate normal distribution.

*Proof.* The asymptotic normality of the set  $\tilde{t}_{ij}$ ,  $i < j$ , is a well known result while the asymptotic normality of the set  $\tilde{V}_{ij}$ ,  $i < j$ , follows from the theory of generalized U statistics. It is thus sufficient to show that in the limit the two sets firstly have the same variances and covariances and secondly that their means differ by a constant factor.

Along the same lines as Mann and Whitney (1947) obtained (cf. also Crouse (1966))

$$\begin{aligned} \text{var } V_{ij} = & n_i^2 n_j^2 \{ \int F_i^2 dF_j - (\int F_i dF_j)^2 \} \\ & + n_i n_j^2 \{ \int F_j^2 dF_i - (\int F_j dF_i)^2 \} + o(N^2) \end{aligned}$$

it can be shown that

$$\text{covar}(V_{ij}, V_{ih}) = n_i n_h n_j \{ \int F_j F_h dF_i - (\int F_j dF_i)(\int F_h dF_i) \} + o(N^2).$$

Under the assumption of continuity, i.e.  $F(x)$  possesses a derivative  $f(x)$  which is everywhere continuous, it readily follows that under (1)

$$\lim \text{var } \tilde{V}_{ij} = 1,$$

$$\lim \text{covar}(\tilde{V}_{ij}, \tilde{V}_{ih}) = \lim \sqrt{n_i n_h / (n_i + n_j)(n_i + n_h)}$$

and the same holds for the set  $\tilde{t}_{ij}$ ,  $i < j$ .

Secondly under (1)

$$\lim E(\tilde{t}_{ij}) = (c_j - c_i) / \sigma \lim \sqrt{n_i n_j / N(n_i + n_j)},$$

while

$$\lim E(\tilde{V}_{ij}) = \lim \sqrt{12 n_i n_j / (n_i + n_j)} \int \{F(x + [c_j - c_i] / \sqrt{N}) - F(x)\} dF(x)$$

and under the assumption of continuity this reduces to

$$\begin{aligned} \lim E(\tilde{V}_{ij}) &= (c_j - c_i) \sqrt{12} \int f^2(x) dx \lim \sqrt{n_i n_j / N(n_i + n_j)} \\ &= \sqrt{12} \sigma \int f^2(x) dx \lim E(\tilde{t}_{ij}) \end{aligned}$$

which completes the proof.

*Corollary 1. Under the hypothesis*

$$F_i(x) = F(x), \quad i=1, \dots, m$$

*the two sets  $\tilde{t}_{ij}, i < j$ , and  $\tilde{V}_{ij}, i < j$ , have asymptotically the same multivariate normal distribution.*

*Corollary 2. Let  $\tilde{t}_{ij}, i < j$ , be based on sample sizes  $n_1, \dots, n_m$ ,*

*with  $N = \sum_{i=1}^m n_i$  and  $\tilde{V}_{ij}, i < j$ , on  $n'_1, \dots, n'_m$  where  $n'_i$  is to the nearest integer a constant ratio of  $n_i$  for all  $i$ . If*

$$\lim n_i / n'_i = 12 \sigma^2 \left( \int f^2(x) dx \right)^2$$

*then under the hypothesis (1)*

$$\lim E(\tilde{t}_{ij}) = \lim E(\tilde{V}_{ij})$$

*and hence the sets  $\tilde{t}_{ij}, i < j$ , and  $\tilde{V}_{ij}, i < j$ , asymptotically have the same multivariate normal distribution.*

### 3. General application

Let  $SS_a$  be the sum of squares due to a certain effect in a parametric analysis of variance table and assume  $SS_a$  to be in the form of a contrast among the cell means  $X_{i.}$ . Consider the statistic  $S = SS_a / \sigma^2$ . Since  $SS_a$  is in the form of a contrast among cell means  $S$  can be expressed as a function of the  $\tilde{t}_{ij}$ 's, say  $S(\tilde{t})$ . Assume, as is the case in general, that under the hypothesis (1)  $S(\tilde{t})$  has

asymptotically a  $\chi^2$  distribution with  $p$  degrees of freedom and noncentrality parameter  $S(\lim E[\tilde{t}])$ .

Consider now as distribution-free test statistic  $S(\tilde{V})$ . From Theorem 1  $S(\tilde{V})$  will also have a  $\chi^2$  distribution in the limit with  $p$  degrees of freedom and noncentrality parameter  $S(\lim E[\tilde{V}])$ . Furthermore from Corollary 2 the asymptotic relative efficiency of the test based on  $S(\tilde{V})$  relative to the test based on  $S(\tilde{t})$  is

$12 \sigma^2 \left( \int f^2(x) dx \right)^2$  which is the same as in the case of two samples. It is also clear that under the hypothesis (1) the class of tests based on  $S(\tilde{V})$  will asymptotically have the same properties, of testing for effects independently of each other, as the parametric class based on  $S(\tilde{t})$ .

#### 4. The one-way classification

In the one-way classification with  $m$  groups the statistic of the form  $S$  is

$$\sum_{i=1}^m n_i (X_{i.} - X_{..})^2 / \sigma^2$$

where  $X_{..} = \frac{1}{N} \sum_{i=1}^m n_i X_{i.}$ . This can be written as

$$\sum_{i < j} n_i n_j (X_{i.} - X_{j.})^2 / N \sigma^2$$

or as

$$S(\tilde{t}) = \sum_{i < j} (n_i + n_j) \tilde{t}_{ij}^2 / N$$

The corresponding distribution-free test statistic in terms of the  $\tilde{V}_{ij}$ 's is thus

$$S(\tilde{V}) = \sum_{i < j} (n_i + n_j) \tilde{V}_{ij}^2 / N$$

which can be written as

$$12 \sum_{i < j} \tilde{V}_{ij}^2 / n_i n_j N$$

and this is the same as the test proposed by Crouse (1966).

#### 5. The two-way classification

In the two-way classification with  $r$  rows,  $c$  columns,  $n$  observations per cell, (i.e.  $m=rc$ ,  $n_i=n$ ,  $i=1, \dots, m$ ,  $N=nrc$ ) and  $X_{hij}$  the  $j$ th observation in the  $i$ th cell of the  $h$ th row, the statistics of the form  $S$  for row, column and interaction effects are

$$S_R = cn \sum_h (X_{h..} - X_{...})^2 / \sigma^2,$$

$$S_C = rn \sum_i (X_{.i.} - X_{...})^2 / \sigma^2,$$

$$S_I = n \sum_h \sum_i (X_{hi.} - X_{h..} - X_{.i.} + X_{...})^2 / \sigma^2,$$

where  $X_{h..}$ ,  $X_{.i.}$  and  $X_{...}$  are the row, column and total means.

These statistics can be written as

$$S_R = cn \sum_{h_1 < h_2} (X_{h_1..} - X_{h_2..})^2 / r \sigma^2,$$

$$= n \sum_{h_1 < h_2} (\sum_i [X_{h_1 i.} - X_{h_2 i.}])^2 / rc \sigma^2,$$

$$S_C = n \sum_{i_1 < i_2} (\sum_h [X_{h i_1.} - X_{h i_2.}])^2 / rc \sigma^2,$$

$$S_I = n \sum_{h_1 < h_2} \sum_{i_1 < i_2} (X_{h_1 i_1.} - X_{h_1 i_2.} - X_{h_2 i_1.} + X_{h_2 i_2.})^2 / rc \sigma^2.$$

Let

$$t_{h_1 i_1, h_2 i_2} = X_{h_1 i_1.} - X_{h_2 i_2.}$$

$$\tilde{t}_{h_1 i_1, h_2 i_2} = \sqrt{n/2\sigma^2} t_{h_1 i_1, h_2 i_2}$$

and similarly

$$v_{h_1 i_1, h_2 i_2} = \sum_{j_1} \sum_{j_2} v(X_{h_1 i_1 j_1}, X_{h_2 i_2 j_2}),$$

$$\tilde{v}_{h_1 i_1, h_2 i_2} = \sqrt{6/n^3} v_{h_1 i_1, h_2 i_2}.$$

In terms of  $\tilde{t}$  the above statistics become

$$S_R(\tilde{t}) = 2_{h_1} \xi_{h_2} \left( \sum_i \tilde{t}_{h_1 i, h_2 i} \right)^2 / rc,$$

$$S_C(\tilde{t}) = 2_{i_1} \xi_{i_2} \left( \sum_h \tilde{t}_{h i_1, h i_2} \right)^2 / rc,$$

$$S_I(\tilde{t}) = 2_{h_1} \xi_{h_2} \sum_{i_1} \xi_{i_2} \left( \tilde{t}_{h_1 i_1, h_1 i_2} - \tilde{t}_{h_2 i_1, h_2 i_2} \right)^2 / rc$$

By replacing  $\tilde{t}$  by  $\tilde{v}$  in these expressions one obtains the corresponding distribution-free test statistics. In terms of  $V$  these statistics can be written as

$$12_{h_1} \xi_{h_2} \left( \sum_i v_{h_1 i, h_2 i} \right)^2 / n^3 rc$$

$$12_{i_1} \xi_{i_2} \left( \sum_h v_{h i_1, h i_2} \right)^2 / n^3 rc,$$

$$12_{h_1} \xi_{h_2} \sum_{i_1} \xi_{i_2} (v_{h_1 i_1, h_1 i_2} - v_{h_2 i_1, h_2 i_2})^2 / n^3 rc.$$

Let  $F_{hi}(x)$  be the distribution function associated with the  $i$ th cell of the  $h$ th row. In the case of a fixed effects model the hypothesis of the form (1) becomes

$$F_{hi}(x) = F(x + c_{hi} / \sqrt{N}),$$

(2)

$$c_{hi} = R_h + C_i + I_{hi}, \text{ for } h=1, \dots, r \text{ and } i=1, \dots, c,$$

where it is assumed that  $\sum_h R_h = \sum_i C_i = \sum_h I_{hi} = \sum_i I_{hi} = 0$ .

Under the hypothesis (2) the above statistics are asymptotically independently distributed as  $\chi^2$  variates with  $(r-1), (c-1), (r-1)(c-1)$  degrees of freedom and noncentrality parameters

$$12 \left( \int f^2(x) dx \right)^2 \sum_{h_1} \xi_{h_2} (R_{h_1} - R_{h_2})^2 / r^2,$$

$$12 \left( \int f^2(x) dx \right)^2 \sum_{i_1} \xi_{i_2} (C_{i_1} - C_{i_2})^2 / c^2,$$

$$12 \left( \int f^2(x) dx \right)^2 \sum_{h_1} \xi_{h_2} \sum_{i_1} \xi_{i_2} (I_{h_1 i_2} - I_{h_1 i_1} - I_{h_2 i_2} + I_{h_2 i_1})^2 / r^2 c^2$$

respectively. These noncentrality parameters can also be expressed in the form

$$12 \left( \int f^2(x) dx \right)^2 \sum_h R_h^2 / r,$$

$$12 \left( \int f^2(x) dx \right)^2 \sum_i C_i^2 / c,$$

$$12 \left( \int f^2(x) dx \right)^2 \sum_h \sum_i I_{hi}^2 / rc.$$

On the other hand under the hypothesis (2) the parametric test statistics  $S_R$ ,  $S_C$  and  $S_I$  are asymptotically distributed as  $\chi^2$  variates with  $(r-1), (c-1), (r-1)(c-1)$  degrees of freedom and noncentrality parameters

$$\sum_h R_h^2 / r\sigma^2, \sum_i C_i^2 / c\sigma^2, \sum_h \sum_i I_{hi}^2 / r c \sigma^2$$

respectively, and this verifies the asymptotic relative efficiency obtained in Section 3 for the special case of a fixed effects complete two-way layout.

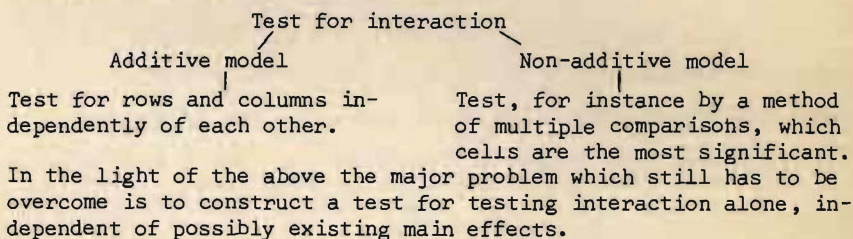
Finally, consider the case where sample sizes are not large and effects not necessarily small. Consider the distribution-free test statistic for rows, namely

$$12 n_1 \sum_{h_2} (\sum_i v_{h_1 i, h_2 i})^2 / n^3 r c.$$

The statistic  $v_{h_1 i, h_2 i}$  measures the departure between two means of

the same column and hence if interaction effects are not present the test statistic for rows is independent of any column effects, and similarly the test statistic for columns is independent of any row effects. Thus if interaction is not present one can test for row and column effects independently of each other. (By this it is not meant that the test statistics are distributed independently of each other because clearly, for small samples, there is some sort of dependence between them. This however is also the case with parametric tests because the error sum of squares enters as a common denominator.) If, however, interaction effects are present, then row, column and interaction effects can not be tested independently of each other.

To evaluate these results consider the minimum requirements for solving a complete two-way layout satisfactorily. Diagrammatically these can be presented as follows.



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